

Mechatronics and the Internet of Things



01 – Introduction

Course Files on GitHub



<https://github.com/ddevoe-umd/enme441>

STEP 1 · INSTALL GIT



macOS

Open the Terminal app and run:

```
xcode-select --install
```

Most Macs already have it — check with `git --version`



Windows

Download Git for Windows and accept the defaults:

git-scm.com/download/win

Then run the commands in the Git Bash terminal it installs.

STEP 2 · DOWNLOAD THE REPOSITORY (ONCE ONLY)

```
# change directory to wherever you keep coursework  
$ git clone https://github.com/ddevoe-umd/enme441.git
```

STEP 3 · GET THE LATEST FILES (ANY TIME)

```
# from inside the ENME441 folder  
$ cd enme441  
$ git pull
```

The repository is public — no GitHub account needed.

Run `git pull` before every lecture. Don't edit files directly in the cloned directory — copy them to another location before editing or the next `git pull` will complain.

COURSE SCHEDULE

Week	Date	Topic	Lab Assignment	
1	31-Aug	Introduction		MECHATRONICS
	2-Sep	Python coding	Lab 1 - Python part 1	
2	7-Sep	cont.		
	9-Sep	ESP32 and Micropython	Lab 2 - Python part 2	
3	14-Sep	cont.	Lab 3 - ESP32 Setup	
	16-Sep	GPIO		
3	21-Sep	cont.	Lab 4 - PWM and IRQs	
	23-Sep	Shift registers	Lab 5 - Shift Registers	
4	28-Sep	cont.	Lab 6 - Cascaded Shift Registers	
	30-Sep	Concurrency		
5	5-Oct	cont.	Lab 7 - Concurrency	
	7-Oct	Stepper Motor Control	Lab 8 - Stepper Motor Control	
6	12-Oct	Fall Break (no class)		
	14-Oct	PCB Design	Lab 9 - PCB Design	
7	19-Oct	Brokered Communication		IoT
	21-Oct	cont.		
8	26-Oct	No class		
	28-Oct	No class		
9	2-Nov	cont.	Lab 10 - Brokered Communication	
	4-Nov	Peer-to-peer Communication		
10	9-Nov	cont.	Lab 11 - Peer-to-Peer Communication	PROJECT
	11-Nov	Final Project		
11	16-Nov	Project work		
	18-Nov	Project work		
12	23-Nov	Project work	Lab 12 – Project Prelim Demo	
	25-Nov	THANKSGIVING BREAK		
13	30-Nov	Project work		
	2-Dec	Project work		
14	7-Dec	Project work		
	9-Dec	Project work		
	16-Dec	Project Competition (final exam time, 4-6pm)		

ENME 441 Course goals

1

Extend your mechatronics knowledge

Strengthen and expand on electronics and mechatronics principles introduced in ENME 350/351.

2

Physical computing with the ESP32

Gain hands-on experience with the ESP32 microcontroller platform.

3

MicroPython

Learn to program embedded hardware (ESP32) in MicroPython.

4

IoT communication

Implement robust broker-based and peer-to-peer messaging.

5

Circuit board development

Learn basic circuit design, PCB layout, and soldering.

6

Design real IoT objects

Design and implement a functional IoT object (Web-enabled laser turret).

Required knowledge base and skill set



Programming

Structured programming basics (ENME 202).



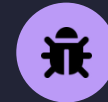
Circuits

Lumped-parameter circuit analysis, op-amp circuits, etc. (ENME 350).



Sensor & actuator interfacing

Voltage buffers, differential amplifiers, bridge circuits, motor control, etc. (ENME 351).



Debugging

The ability to identify and correct errors in both circuits and code.



An attitude of exploration

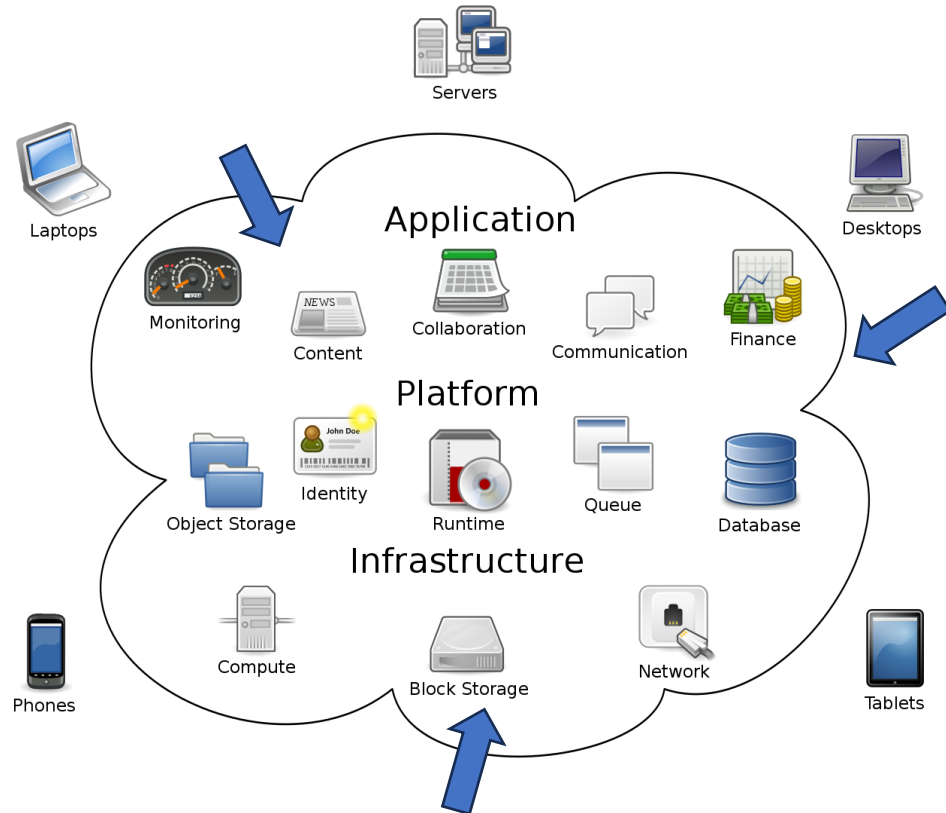
A desire to explore, play around, and leverage online resources including LLMs.

01

Cloud, Edge, and IoT



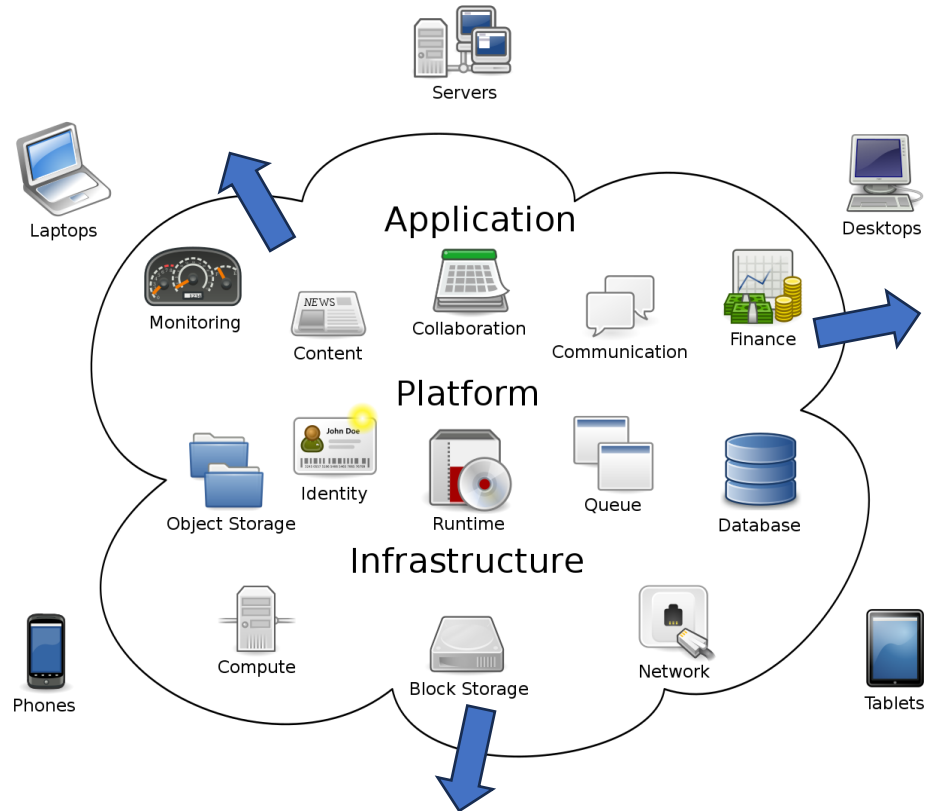
Cloud Computing



Move computation & storage inward → leverage centralized (cloud) resources

- **Centralized compute and storage**
Servers, storage and analytics live in one elastic pool that any device can reach.
- **Thin endpoints**
Devices mostly report to the cloud; heavy computation happens elsewhere.
- **Costs move to the network**
Add data (e.g. sensor measurements) must be shipped upstream — bandwidth and latency are the price.

Edge Computing



Move computation outward into edge nodes

- **Compute at the source**
Filtering, thresholding and control decisions happen at the nodes.
- **Faster communications**
Only send what matters. Raw data stays local while summaries and events go upstream.
- **Works offline**
Local loops keep running when the link to the cloud is slow or gone.
- **Improved data security**
Can avoid the need to send sensitive data over the network

The Internet of Things

IoT objects can operate within both cloud computing and edge computing frameworks. The choice depends on:

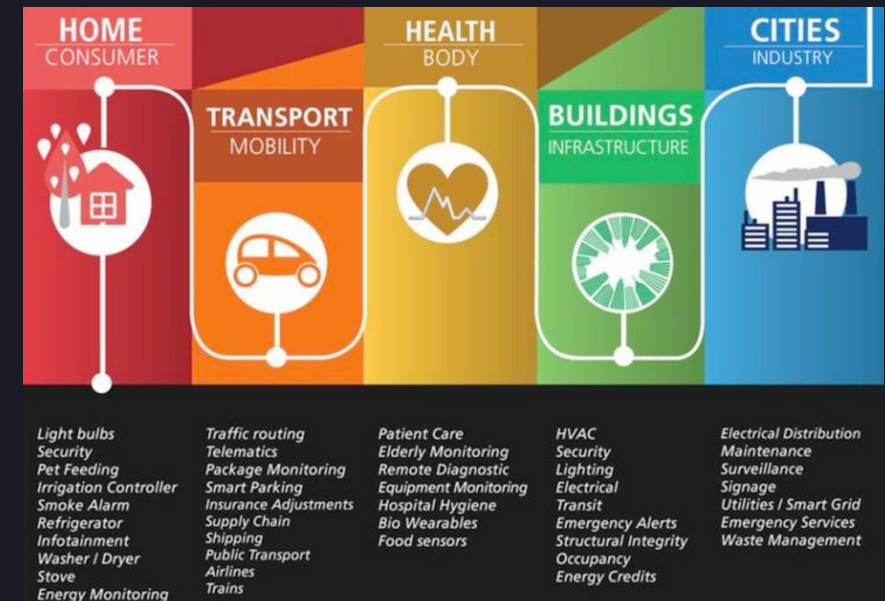
- Computational power available at the node (really a question of IoT object cost)
- Network bandwidth limits
- Data security requirements



“

“In the next century, planet earth will don an electronic skin. It will use the Internet as a scaffold to support and transmit its sensations. This skin is already being stitched together. It consists of millions of embedded electronic measuring devices: thermostats, pressure gauges, pollution detectors, cameras, microphones, glucose sensors, EKGs, electroencephalographs. These will probe and monitor cities and endangered species, the atmosphere, our ships, highways and fleets of trucks, our conversations, our bodies – even our dreams.”

BusinessWeek, 1999



02

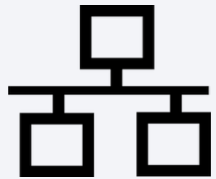
connectivity

How IoT objects talk



Communication platforms

Just a sampling of currently-prevalent communication platforms. There are many more!



Ethernet

IEEE 802.3 · wired LAN



Wi-Fi

Wireless Fidelity · 802.11



Bluetooth

Short-range personal area network



Zigbee

Low-power, low-rate mesh



NFC

Near-field communication



RFID

Radio frequency identification



LoRaWAN

Long Range wide area network



GPRS

General Packet Radio Service



Cellular

GSM · 3G · LTE (4G) · 5G

Communication protocols

Communication protocols define how messages are structured, addressed, and delivered over the communication platform



MQTT

Message Queue
Telemetry Transport



AMQP

Advanced Message
Queuing Protocol



CoAP

Constrained
Application Protocol



DDS

Data Distribution
Service



XMPP

eXtensible Messaging
and Presence Protocol



REST

Representational
State Transfer



JMS

Java Message
Service



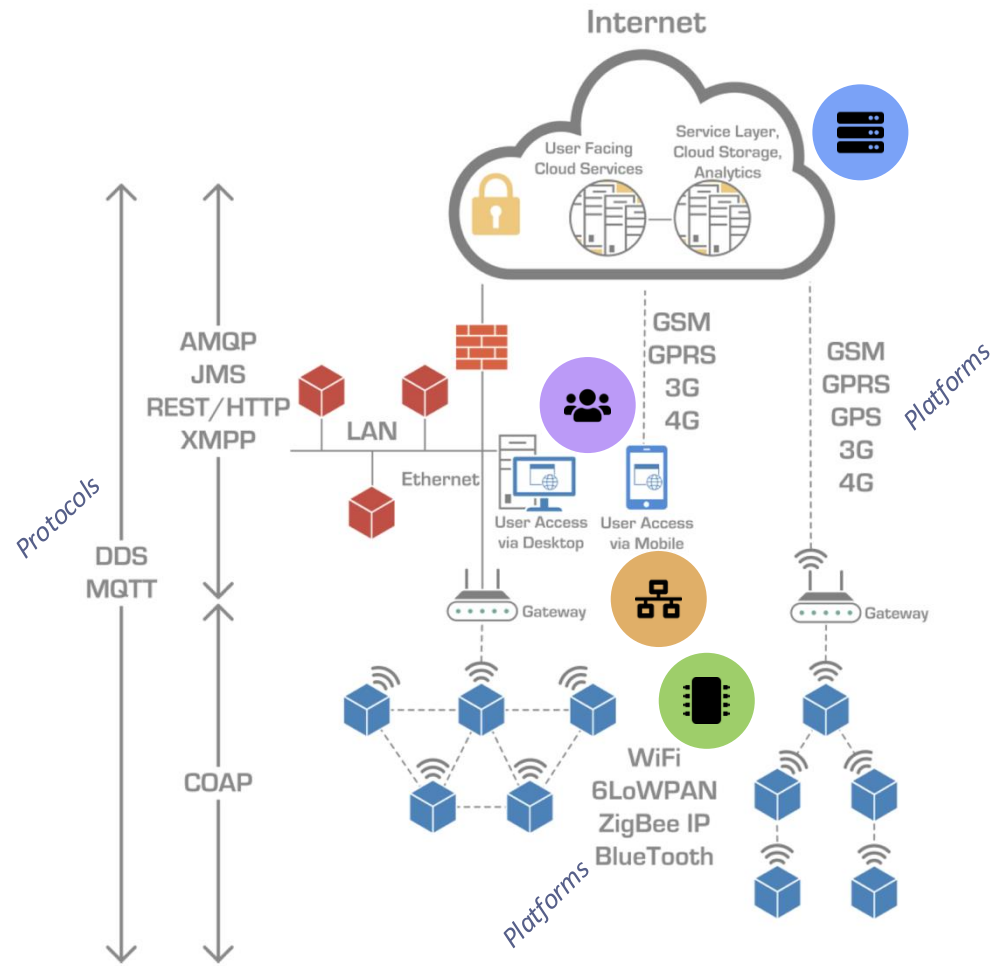
Etc...

and a long tail
of others

In ENME441 we will use:

- *MQTT for broker-based messaging*
- *REST protocols (GET/POST) for peer-to-peer messaging*

The IoT stack



*"Messaging Technologies for the Industrial Internet and the Internet of Things,"
ADLINK Whitepaper, 2017*



Edge Devices

Microcontrollers · board-level computers · embedded systems



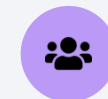
Gateways

Bridge the local wireless links onto IP networks



Data System (Cloud)

Storage, analytics and the user-facing service layer



Users

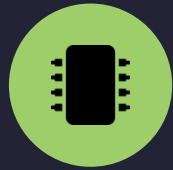
Desktop and mobile access to everything below

Next steps



Python (review)

Data types, flow control, functions, objects.



ESP32 microcontrollers

Microcontroller with input/output pins and multiple communication options



MicroPython

Python on “bare metal”

Bring a laptop with Python installed to the next lecture.